

1) Levels of Testing

Testing levels define at what stage of development the software is tested.

There are 4 main levels.

- 1) Acceptance Testing
- 2) System Testing
- 3) Integration Testing
- 4) Unit Testing

1) Unit Testing

tests the smallest piece of code, a single function or method.

eg: testing a `calculateTax()` in

2) Integration Testing

checks whether two or more modules communicate correctly when combined.

eg: after login module and database module and database module are built separately.

3) System Testing

tests the full application as whole system. - all modules together.

4) Acceptance Testing (UAT)

is done by the client or end user to confirm the software meets their business needs.

eg: the bank's manager uses the app and approves it before going live.

2) i) Functional Testing
Test what the system does.

Ex: 1. Smoke Testing

i) Quick check, basic features work. eg: Can user login and open homepage.

2. Regression Testing:-

Re-test after code changes.

Ex. old features still work after new build.

ii) Non-functional Testing:-

Tests How well the system works,

eg: 1) Performance Testing:-

check speed and response time.

Ex: APP loads in under 3 seconds.

2) Security Testing:-

Detect vulnerabilities and breaches.

Ex. i) No. SQL injection in login page.

ii) checking whether a bank's login page is protected against SQL injection attacks.